

40.520 Stochastic Models

Poisson Processes

Xiaotang Yang

Engineering Systems and Design (ESD)
Singapore University of Technology and Design (SUTD)

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Outline

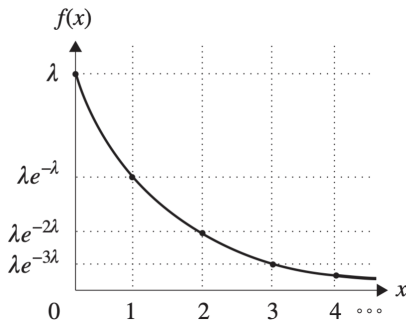
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The Exponential Distribution

Definition

- A continuous random variable X is said to have an **exponential distribution** with parameter λ : $X \sim \text{Exp}(\lambda)$ if its probability density function is given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0; \\ 0, & x < 0. \end{cases}$$



CDF, Mean and Variance

- The CDF of an exponentially distributed random variable is given by

$$F(x) = \int_{-\infty}^x f(u)du = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0; \\ 0, & x < 0. \end{cases}$$

- The mean and variance of an exponentially distributed random variable are
 - Mean: $E[X] = \frac{1}{\lambda}$
 - Variance: $Var[X] = \frac{1}{\lambda^2}$

The Memory-Less Property

- We can show that

$$P(X > s + t \mid X > t) = P(X > s)$$

for a random variable with exponential distribution.

- If we think of X as being the lifetime of some instrument, then this property states that the instrument does not remember that it has already been in use for a time t .

The Memory-Less Property

- The only continuous distribution to have this property is the exponential distribution.

Example: Service Time in a Bank

- Suppose that the amount of time one spends in a bank is exponentially distributed with mean 10 minutes, that is, $\lambda = 0.1$.
- What is the probability that a customer will spend more than 5 minutes in the bank?
- What is the probability that a customer will spend more than 15 minutes in the bank, given that she is still in the bank after 10 minutes?

Several Further Properties of Exponential Distribution

Suppose that $\{X_1, X_2, \dots, X_n\}$ are independent exponential random variables, with X_i having rate λ_i , $i = 1, 2, \dots, n$.

The distribution of

$$Y = \min\{X_1, X_2, \dots, X_n\}$$

follows an exponential distribution with a rate $\sum_{i=1}^n \lambda_i$. That is,

$$P(Y > x) = e^{(-\sum_{i=1}^n \lambda_i)x}.$$

Several Further Properties of Exponential Distribution

Suppose that X_1 and X_2 are independent exponential random variables with rates λ_1 and λ_2 , respectively. The probability

$$P(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Example: Post Office

- Consider a post office that is run by two clerks. Suppose that when Mr. Smith enters the post office he discovers that Mr. Jones is being served by one of the clerks and Mr. Brown by the other. Suppose also that Mr. Smith is told that his service will begin as soon as either Jones or Brown leaves.
- If the amount of service time for each clerk is exponentially distributed with rate λ , what is the probability that, of the three customers, Mr. Smith is the last one to leave?

Example: Post Office Again

- Suppose that you arrive at a post office having two clerks at a moment when both are busy but there is no one else waiting in line. You will enter service when either clerk becomes free.
- If service times for clerk i are exponential with rate λ_i , $i = 1, 2$, find $E[T]$, where T is the amount of time that you spend in the post office.

Several Further Properties of Exponential Distribution

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed exponential random variables having mean $1/\lambda$. The sum

$$Z_n = X_1 + X_2 + \dots + X_n$$

has a gamma distribution; that is, its PDF is given by

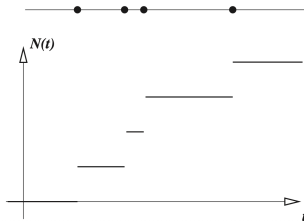
$$f_{Z_n}(t) = \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!}.$$

The Poisson Process

Counting Processes

A stochastic process $\{N(t), t \geq 0\}$ is said to be a *counting process* if $N(t)$ represents the total number of “events” that have occurred up to time t . Hence, a counting process $N(t)$ must satisfy:

- (i) $N(t) \geq 0$.
- (ii) $N(t)$ is integer-valued.
- (iii) If $s < t$, then $N(s) \leq N(t)$.
- (iv) For $s < t$, $N(t) - N(s)$ equals the number of events that have occurred in the interval $(s, t]$.



Counting Processes

- A counting process is said to possess **independent increments** if the numbers of events that occur in disjoint time intervals are independent.
- A counting process is said to possess **stationary increments** if the distribution of the number of events that occur in any interval of time depends only on the length of the time interval.

Construction of a Poisson Process

- Begin with a sequence $\tau_1, \tau_2, \dots$ of independent exponential random variables, all with the same mean $1/\lambda$.
- We build a model in which “events” occur from time to time. In particular,
 - the first event occurs at time τ_1 ;
 - the second event occurs τ_2 time units after the first event;
 - ...

Construction of a Poisson Process

- The random variables $\{\tau_k\}_{k \in \{1,2,\dots\}}$ are called the inter-arrival times.
- The waiting time of the n -th event is

$$S_n = \sum_{k=1}^n \tau_k.$$

- A process $\{N_t, t \geq 0\}$ is said to be a Poisson process if it counts the number of events in the above model that occur at or before time t .

Distribution of Poisson Processes

- We have

$$P(N_t = k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}, \quad k = 0, 1, 2, \dots$$

i.e., N_t follows a Poisson distribution.

Mean and Variance of Poisson Increments

Mean:

$$E[N_t - N_s] = \lambda(t - s).$$

Variance:

$$\text{Var}[N_t - N_s] = \lambda(t - s).$$

Poisson Process as a Counting Process

- In summary, the Poisson process must satisfy:
 - $N_t \geq 0$;
 - N_t is integer valued;
 - $N_t \geq N_s$ for $t \geq s$;
 - $N_t - N_s$, the number of events that occur in the interval $(s, t]$, follows a Poisson distribution.
- The Poisson process is a special case of counting processes.

Poisson Process

Definition 1 of the Poisson Process

A Poisson process having rate λ is a counting process such that

1. $N(0) = 0$.
2. The process has independent increments.
3. The number of events in any interval of length t is Poisson distributed with mean λt .
That is, $\forall s, t \geq 0$,

$$\mathbb{P}\{N(t+s) - N(s) = n\} = \frac{e^{-\lambda t}(\lambda t)^n}{n!}, \quad n = 0, 1, \dots$$

Poisson Process

Definition 2 of the Poisson Process

A Poisson process having rate λ is a counting process such that

1. $N(0) = 0$.
2. The process has stationary and independent increments.
3. $\mathbb{P}\{N(\delta) = 1\} = \lambda\delta + o(\delta)$.
4. $\mathbb{P}\{N(\delta) \geq 2\} = o(\delta)$.

Remark: The function f is said to be $o(h)$ if $\lim_{h \rightarrow 0} \frac{f(h)}{h} = 0$.

Exercise

Show equivalence between Definitions 1 and 2 of the Poisson process.

Example: Immigration

- Suppose that people immigrate into a territory at a Poisson rate $\lambda = 1$ per day.
- What is the expected time until the 10th immigrant arrives?
- What is the probability that the elapsed time between the 10th and the 11th immigrants exceeds 2 days?